

$$A = \{1, 2, 3\}$$

$$R = \{(1, 1), (2, 2), (1, 2)\}$$

Is it reflexive? **No**

$$R = \{(a, b) \mid a, b \text{ play for the same team}\}$$

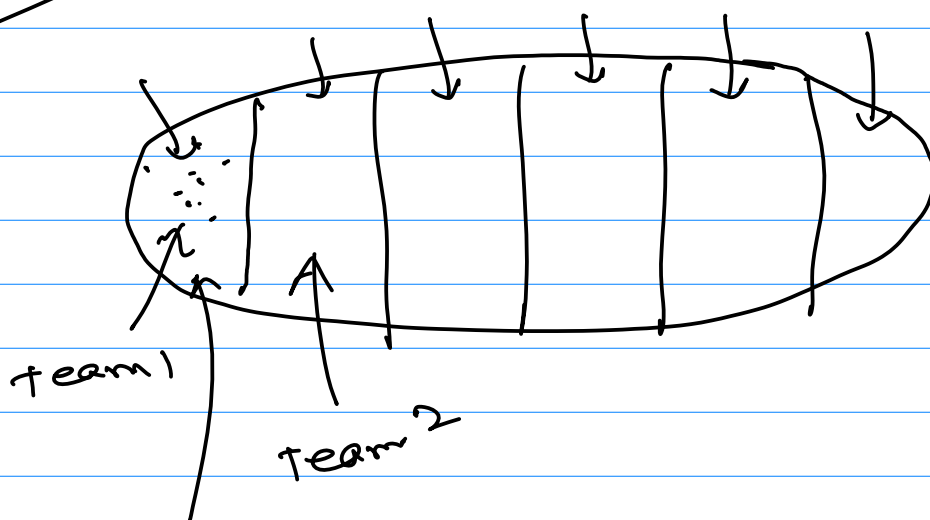
Is it reflexive? **Yes** $\rightarrow (a, a) \in R$

Is it symmetric? **Yes**

Is it transitive? **Yes**

Rohit Sharma

equivalent relation.



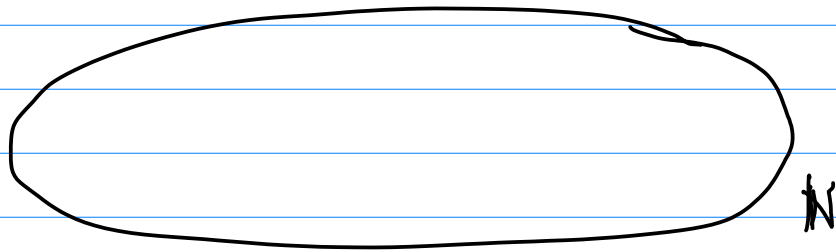
$$\text{Team 1} = \{\text{Rohit, Virat, ...}\}$$

Equivalent classes

Babbar Azam

Team 2 = { Babbar, ... }
↘

N



dividing by 4

What are the possible remainders.

0, 1, 2, 3

$$R = \{ (a, b) \mid a \overset{\text{congruence}}{\equiv} b \pmod{4} \}$$

$$\Downarrow \boxed{4 \mid a - b}$$

4 divides a-b.

$$\begin{array}{c} 5 \\ \uparrow \end{array} \equiv \begin{array}{c} 1 \\ \uparrow \end{array} \pmod{4}$$

$$5 - 1 = 4$$

$$4 \mid 4$$

$$4 \mid (7 - 3)$$

$$7 \equiv 3 \pmod{4}$$

$$4 \mid 19 - 3$$

$$\underline{19} \equiv \underline{3} \pmod{\underline{4}}$$

$$19 \equiv 3 \pmod{4}$$

$$19 \equiv (\pmod{4})$$

modulo 4

$$\textcircled{X} R = \{ (a, b) \mid 4 \text{ divides } (a-b) \} \quad \checkmark$$

Reflexive . . . Yes

(a, a) 4 divides 0

Symmetric Yes

4 divides $a-b$

4 divides $b-a = -(a-b)$

Transitive Yes

$(a, b) \in R \Rightarrow 4 \text{ divides } (a-b) \quad \checkmark$

$(b, c) \in R \Rightarrow 4 \text{ divides } (b-c) \quad \checkmark$

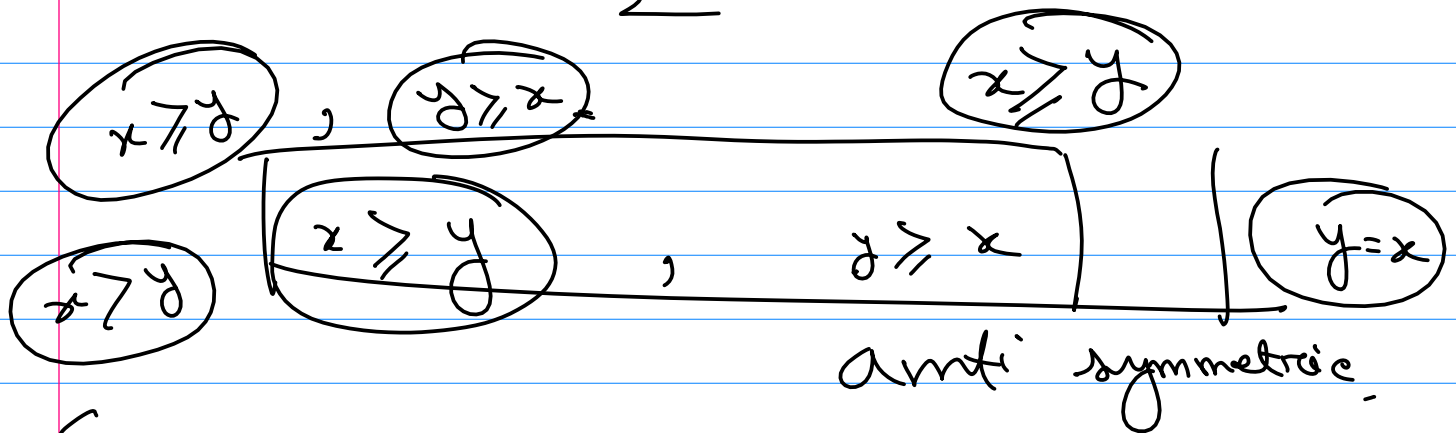
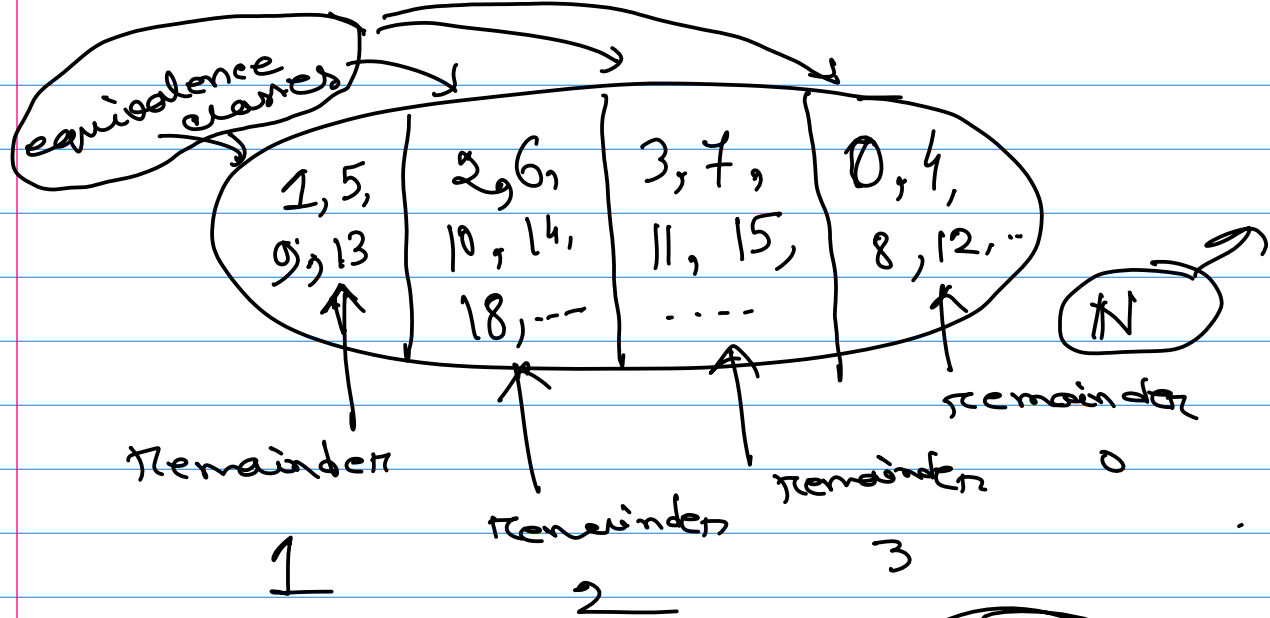
Is $(a, c) \in R$?

Yes

$(a, c) \in R$

4 divides $(a-b) + (b-c)$
 $= (a-c)$

$R = \{(a, b) \mid 4 \text{ divides } (a-b)\} \rightarrow \text{equivalence relation}$



anti symmetric { a gets more or equal than b
b gets more or equal than a
a and b must get equal.

a gets more or equal to b
 b gets more or equal to c
 a gets more or equal to c

$$(s+1, s+2) \in R$$

$$A = \left\{ \underset{\uparrow}{x} ; x \in \mathbb{R} \right\}$$

$$R = \left\{ \left(\underline{a}, \underline{b} \right) \mid \text{---} \text{---} \text{---} \text{---} \right\}$$

$$(\underline{x}, \underline{y}) \in R$$

$$3 \geq 2$$

$$2 \geq 3 \quad (\times)$$

$$\textcircled{2} \geq \textcircled{2}$$

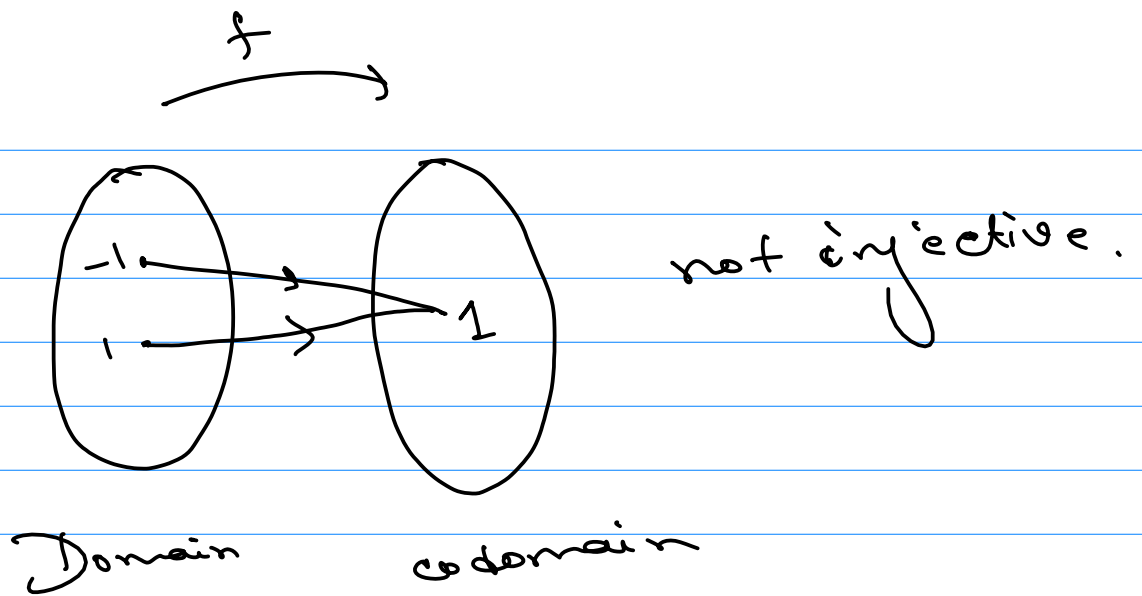
$$\textcircled{a \neq b} \quad (a, b) \in R$$

$$(b, a) \notin R$$

$$\textcircled{x \geq y}$$

$$\textcircled{y \not\geq x}$$

$$\underline{x \neq y}$$



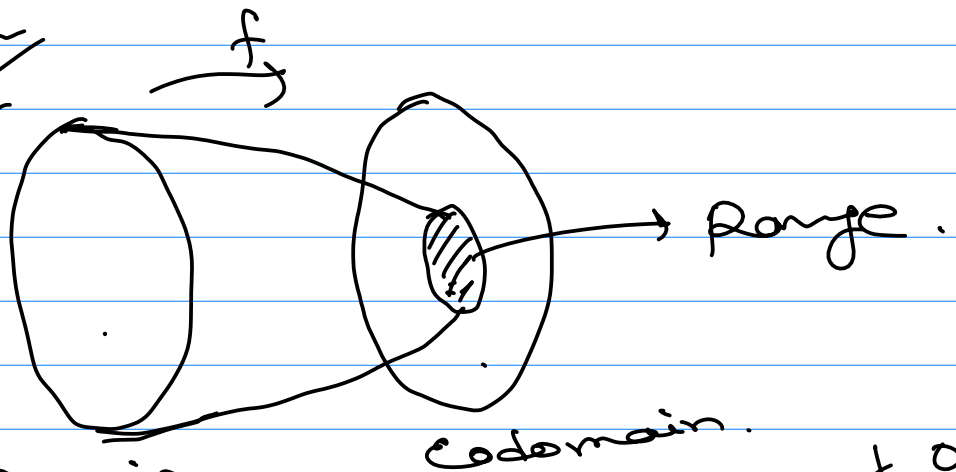
$$\left. \begin{aligned} f: \mathbb{R} &\rightarrow \mathbb{R} \\ f(x) &= x^2 \end{aligned} \right\}$$

$$f(-1) = 1$$

$$f(1) = 1$$

not injective

Real valued
function



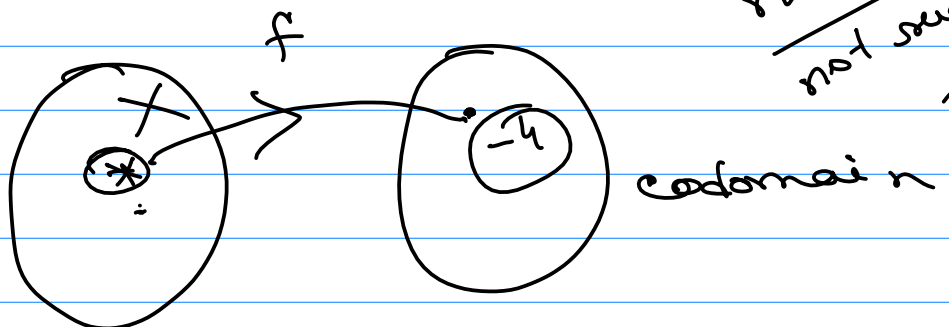
$f(x) = x^2$

Range(f)

$$= \{x \mid x \geq 0, x \in \mathbb{R}\}$$

Domain

No



not onto
not surjective

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = 2x + 3$$

onto?

a, b

$$f(a) = f(b)$$

$$\Rightarrow 2a + 3 = 2b + 3$$

$$\Rightarrow 2a = 2b$$

$$\Rightarrow a = b$$

one to one

Does there exist some input x for which $f(x) = m$?

$$f(x) = x^2$$

not one to one

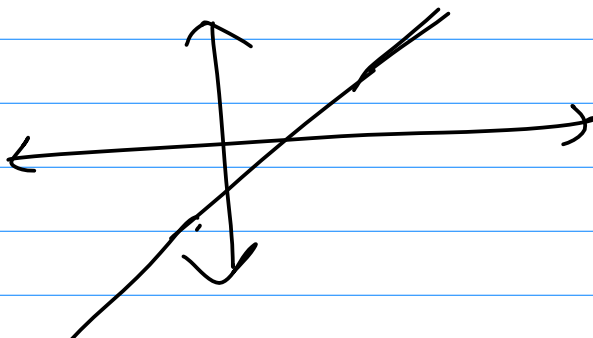
$$f(x) = m$$

$$\Rightarrow 2x + 3 = m$$

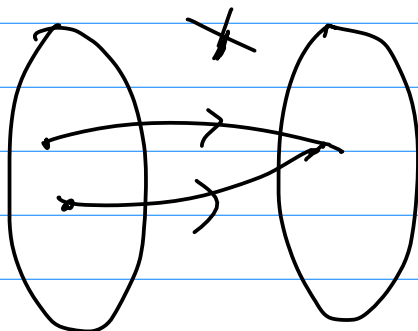
$$\Rightarrow x = \frac{m-3}{2} \in \mathbb{R}$$

onto

$$f\left(\frac{m-3}{2}\right) = m$$



$f: \mathbb{R} \rightarrow \mathbb{R}$
 $f(x) = x^2$
 not onto

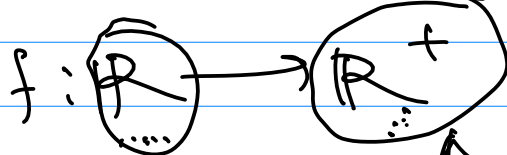


cannot be injective

$$f(x) = x^2$$

$$f(-1) = 1$$

$$f(1) = 1$$



$$f(x) = x^2$$

Range (f)

$$= \{x \mid x \geq 0, x \in \mathbb{R}\}$$

onto

codomain (f)

$\sqrt{a}$

$$2 \in \mathbb{R}^+$$

$$f(\sqrt{2}) = 2$$

$$\sqrt{2} \in \mathbb{R}$$

$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = |x+4|$$

$$\left. \begin{aligned} f(-6) &= 2 \\ f(-2) &= 2 \end{aligned} \right\}$$

not injective

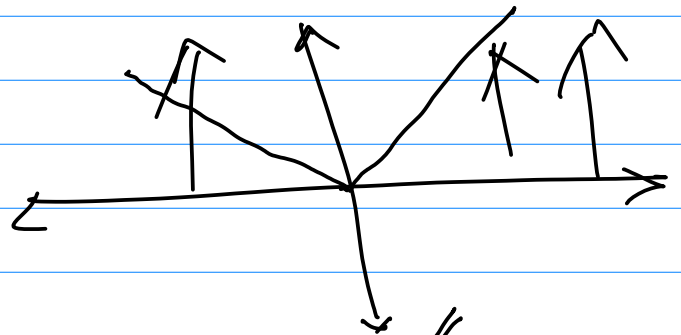
$$f(x) = |x|$$

$$f(x) = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$f(-4) = |-4+4| = 0$$

$$\text{Range}(f) = \mathbb{R}^+ = [0, \infty)$$

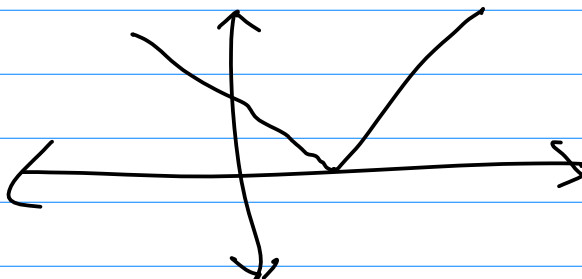
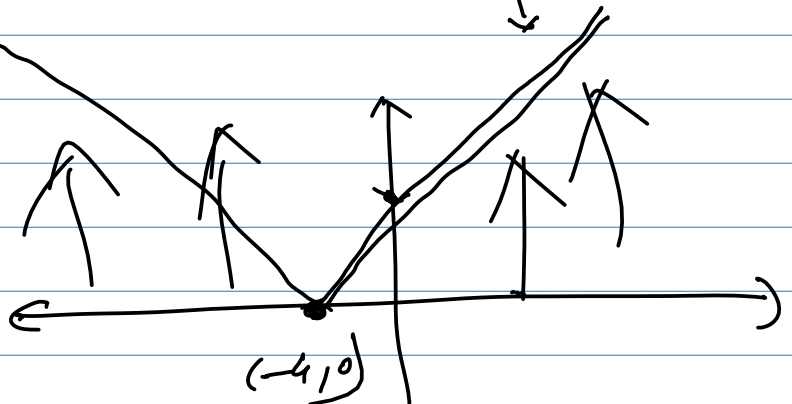
$$f(x) = |x| \rightarrow$$



$$|x+4| = 0$$

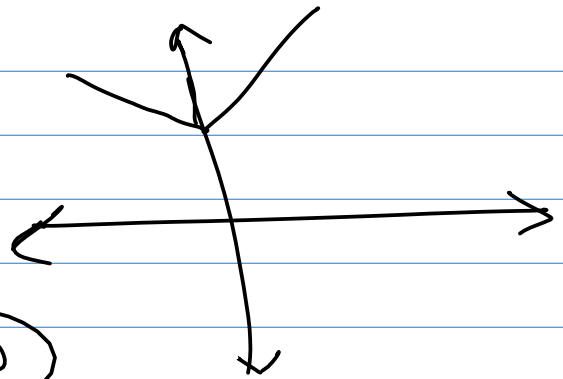
$$f(x) = |x+4|$$

$$f(x) = |x-4|$$



$$f(x) = |x| + 4$$

$$\text{Range}(f) = [4, \infty)$$



$$f(x) = \frac{x+2}{\sqrt{4x-8}}$$

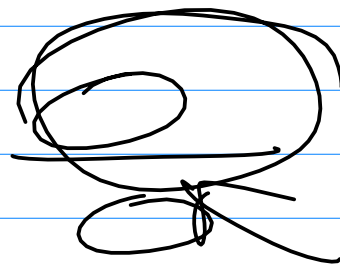
$$4x - 8 = 0$$

$$x = 2$$

Domain

$$\mathbb{R} \setminus \{2\}$$

$$\sqrt{}$$



$$f(x) = \sqrt{x-2} \quad [2, \infty)$$

$$x - 2 \geq 0$$

$$x \geq 2$$

$$f: \mathbb{N} \rightarrow \mathbb{N}$$

$$f(n) = \begin{cases} 2n & n \text{ is odd} \\ \frac{n}{2} & n \text{ is even} \end{cases}$$

onto but not one to one

$$\left. \begin{aligned} f(1) &= 2 \\ f(4) &= 2 \end{aligned} \right\}$$

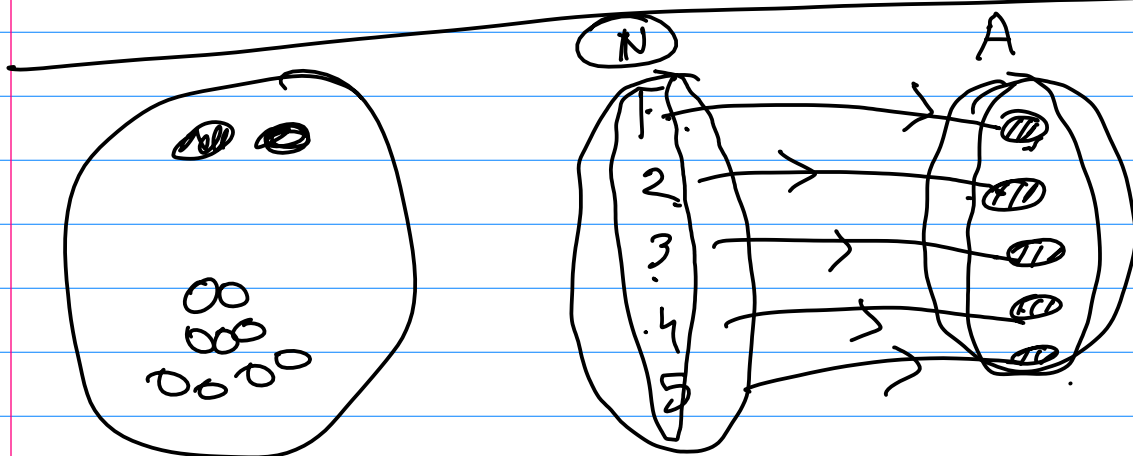
not injective

$$\cancel{2k}$$

$$\cancel{2k+1}$$

$$f\left(\frac{2(2k+1)}{2}\right) \xrightarrow{\text{even}} \frac{2(2k+1)}{2}$$

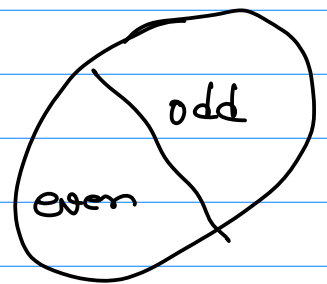
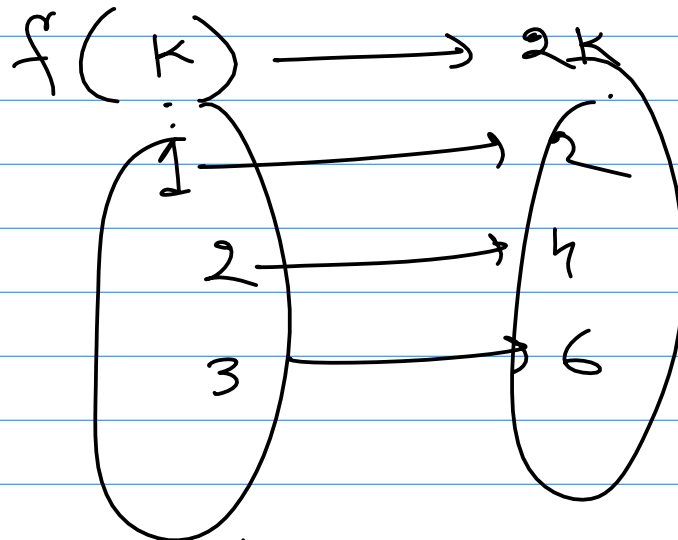
$$f\left(\frac{2(2k+1)}{2}\right) = \frac{2(2k+1)}{2} = 2k+1$$



$A = \text{Set of even numbers } \{2n \mid n \in \mathbb{R}\}$

$B = \text{Set of odd numbers } \{2n+1 \mid n \in \mathbb{R}\}$

$$B \longrightarrow A$$



$$|A| = |B| = |\mathbb{N}|$$

$$f: \mathbb{N} \longrightarrow A$$

$$X = \{a, b, c, d, e\}$$

$$A = \{ \underline{(a, b)}, \underline{(b, c)}, (a, c), (b, d), (d, e) \}$$

not
transitive

$$(b, c) \notin A$$

$$R_1 = \left\{ \begin{array}{l} (\text{Jan}, \text{Feb}), (\text{Jan}, \text{Jan}), (\text{Feb}, \text{Feb}), (\text{Feb}, \text{Jan}) \\ (\text{Sep}, \text{Nov}), (\text{Sep}, \text{Sep}), (\text{Sep}, \text{Dec}) \\ (\text{Nov}, \dots) \end{array} \right\}$$

$$R_2 = \left\{ (\text{Jan}, \text{Jan}), (\text{Jan}, \text{Mar}), \dots \right\}$$

$$R_3 = \left\{ (a, c) \mid \underline{(a, b)} \in R_1, (b, c) \in R_2 \right\}$$

$(\text{Feb}, \text{Jan}) \in R_1, \quad (\text{Jan}, \text{Mar}) \in R_2$
 $(\text{Feb}, \text{Mar}) \in R_3$

$$f(x) = \frac{x}{1+x^2}$$

$$f(x) = 1+x^2$$

$$f(x) = \frac{1}{1+x^2}$$

$$f(x) = x$$